17 CHROMOSOMES, GENES & PROTEINS 21/01/23

Chromosomes:-

made of DNA which contains genetic information in the form of genes.
- gene is a length of DNA that codes for a protein
  - allele is an alternative form of a gene

XX & XY chromosomes:-

- sex is determined by an entire chromosome pair
  - females have the sex chromosomes 'XX'
  - males have the sex chromosomes 'XY'
- only the father can pass on a Y chromosome therefore he is responsible for determining the sex of the child.
  - he produces 250 million sperm cells during intercourse
    - half of those carry the X chromosome
      - if it fertilises the egg, the fetus = female
    - half of those carry the Y chromosome
      - if it fertilises the egg, the fetus = male

- sequences of bases in a gene determines the sequence of amino acids used to make a specific protein.
- different sequences of amino acids give different shapes/functions to protein molecules.

(mother)

gametes

(father)

	X	X
X	XX	XX
Y	XY	XY

- DNA controls cell function by controlling the production of proteins, including enzymes, membrane carriers and receptors for neurotransmitters.

How is a protein made:

- the gene coding for the protein remains in the nucleus
- messenger RNA (mRNA) is a copy of a gene
- mRNA molecules are made in the nucleus and move to the cytoplasm
- the mRNA passes through ribosomes
- the ribosome assembles amino acids into protein molecules
- the specific sequence of amino acids is determined by the sequence of bases in the mRNA

- most body cells in an organism contain the same genes, but many genes in a particular cell are not expressed because the cell only makes the specific proteins it needs

- a haploid nucleus contains a single set of chromosomes
- a diploid nucleus contains two sets of chromosomes
  - in a diploid cell, there is a pair of each type of chromosome
  - in a human diploid cell there are 23 pairs

MITOSIS 21/01/23

Mitosis:-

- a nuclear division giving rise to genetically identical cells
- mitosis is used for: growth, repair of damaged tissues & asexual reproduction.
- when cells divide (mitosis), their chromosomes double beforehand
  - during mitosis, the copies of chromosomes separate, maintaing the chromosome number in each daughter cell
- stem cells are unspecialised cells that divide by mitosis to produce daughter cells that can become specialised for specific functions

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MEIOSIS 21/01/23

Meiosis:-

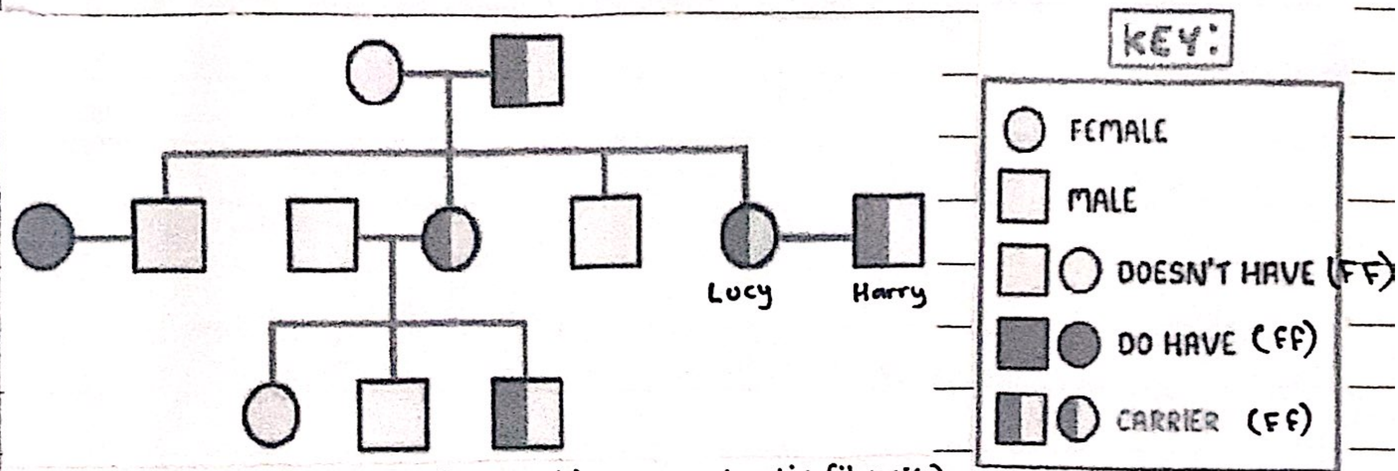
- involved in the production of gametes
- a reduction division in which the chromosome number is halved from diploid to haploid resulting in genetically different cells.

MONOHYBRID INHERITANCE 21/01/23

Inheritance:-

- the transmission of genetic information from generation to generation
- genotype is the combination of alleles that control each characteristic
- phenotype is the observable characteristics of an organism.
- homozygous is having 2 identical alleles of a gene
  - two identical homozygous individuals that breed together will be pure-breeding.

- heterozygous is having two different alleles of a particular gene
  - └ a heterozygous individual will not be pure-breeding.
- a dominant allele is the allele that is expressed if it is present in the genotype
- a recessive allele is the allele that is only expressed when there is no dominant allele of the gene present in the genotype



⇒ Pedigree diagram (cystic fibrosis)

what are the chances of Lucy & Harry's baby having cystic fibrosis?

(L)

(H)		F	f
	F	FF	Ff
	f	Ff	ff

← 1/4 chance

- codominance is a situation in which both alleles in heterozygous organisms contribute to the phenotype

genotype	phenotype
I^AI^A OR I^AI^o	A
I^BI^B OR I^BI^o	B
I^AI^B	AB
I^oI^o	O